How a classifier decides

AI · Level 3 · Lesson 2



Today we will

- Describe each example by features: numbers you can measure.
- Classify a new example by its nearest neighbours, and see how the number of neighbours changes the answer.
- Explain why the choice of features matters.

Remember

- Features are numbers you can measure, like softness from 0 to 10.
- A new example takes the class most of its k nearest neighbours have. With k = 1 one example decides; a huge k loses the details.
- Good features separate the classes: softness and colour do, weight alone doesn’t.

Neighbours on paper

1. Measure (5 min): eight leaves from two kinds of tree.
2. Chart (5 min): plot them as labelled dots.
3. Mystery (5 min): a partner measures a new leaf.
4. Vote (5 min): ask 1, 3 and 5 neighbours.
5. Swap features (5 min): try a poor feature instead.

Works in pairs or at home. Any two kinds of objects will do: leaves from two trees, two kinds of beans, or two shapes you draw. Use a ruler for every measurement.

Let's discuss

1. Why might k = 1 give a strange answer?
2. Which two features would you use to sort bananas into ripe and not ripe?
3. Can a classifier be better than the features it is given? Why?

Before you go

- One feature I would measure for a classifier of my own, and why.